Unit 2 Quiz B: Coaching Guide

Don't just check the letter. For each question, answer the 'Ask yourself' prompts BEFORE you read the steps.

Q	1	2	3	4	5	6	7	8	9	10	11	12	13
Key	A	C	B	C	C	B	B	B	A	C	A	A	B

- **Score it:** right + sure = solid. Right + guess = review. Wrong = full walkthrough + Go deeper.
- Re-attempt every miss on paper tomorrow, without looking. If you get it right then, it sticks.

Question 1

2.1 Center of mass (2-D)

Three particles: 1.0 kg at (0, 0), 1.0 kg at (4.0 m, 0), and 2.0 kg at (0, 3.0 m). Where is the center of mass?

ASK YOURSELF FIRST

- Handle x and y completely separately. Which particles contribute to x? Which to y?
- The heaviest particle sits on the y-axis. Should y_{cm} be pulled up toward it?

STEP BY STEP

1. x_{cm} = (1.0·0 + 1.0·4.0 + 2.0·0) / 4.0 = 4.0/4.0 = 1.0 m.
2. y_{cm} = (1.0·0 + 1.0·0 + 2.0·3.0) / 4.0 = 6.0/4.0 = 1.5 m.
3. So (1.0 m, 1.5 m).

ANSWER A. (1.0 m, 1.5 m)

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** Averaged the positions without weighting by mass: x = 4/3, y = 3/3.
- **C:** Divided x by 2 instead of by the total mass of 4.0 kg.

GO DEEPER Sketch the three points and mark your answer. Does it sit inside the triangle, nearer the 2 kg particle? A sketch catches most slips.

Question 2

2.1 Systems

A person stands at one end of a boat floating at rest on still water (ignore water friction). The person walks to the other end. What happens?

ASK YOURSELF FIRST

- Pick the system: person + boat. Are there any HORIZONTAL forces from outside the system?
- If the net external force is zero and everything started at rest, can the system's center of mass move?

STEP BY STEP

1. System = person + boat. Horizontally, no external force acts (no water friction).
2. Net external force = 0, so the center of mass doesn't accelerate. It started at rest, so it stays put.
3. The person moves forward. Internally, the boat pushes the person forward and the person pushes the boat back (3rd law).
4. To keep the center of mass fixed, the boat must slide backward.

ANSWER C. The center of mass stays put and the boat slides backward

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Internal forces can't move the center of mass, but they CAN move the parts of the system relative to each other.
- **B:** Only an external force can move the center of mass.

GO DEEPER Person 60 kg, boat 120 kg, and the person walks 3.0 m relative to the water. How far does the boat move? (60 × 3.0 = 120 × d, so d = 1.5 m backward.)

Question 3

2.2 Free-body diagrams

A ball is thrown at an angle. At the HIGHEST point of its path (air resistance negligible), which forces act on it?

ASK YOURSELF FIRST

- After the ball leaves the hand, what objects touch it or pull on it?
- Does it need a force to KEEP moving horizontally? (Think of Newton's 1st law.)

STEP BY STEP

1. Contact with the hand has ended, so the throw's force is gone.
2. The only interaction left is Earth's gravity: F_g = mg, downward.
3. The horizontal velocity carries on with no force needed (1st law). That's why it's constant in projectile motion.

ANSWER B. Only gravity, downward

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** A 'force of motion'. Motion doesn't need a force, and there's no object to supply one.
- **C:** At the top, $v_y = 0$ but $v_x \neq 0$, and gravity is still on. A ball with no forces would never come down.
- **D:** Forces don't get 'stored' in an object.

GO DEEPER Sketch a_y vs. t for the whole flight. (A flat line at −9.8 m/s², including the top.)

Question 4

2.3 Newton's 3rd law

A heavy truck pushes a small car with a dead battery. They speed up together. How does the force of the truck on the car compare with the force of the car on the truck?

ASK YOURSELF FIRST

- Does the 3rd law have any exceptions for accelerating objects?
- If the forces between them are equal, what makes the truck + car speed up?

STEP BY STEP

1. Truck-on-car and car-on-truck are a 3rd-law pair, so they're always equal and opposite, even while accelerating.
2. Why they speed up: on the CAR, the truck's push is bigger than the friction or drag on the car. On the TRUCK, the road's forward push on its tires is bigger than the car's push back.

3. Different objects, different FBDs. The pair never appears on the same diagram.

ANSWER C. Equal at all times

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Mixes up the pair with the net force on one object.
- **B:** The 3rd law has no 'only when balanced' clause. You're thinking of the 1st law.

GO DEEPER Name the external force that accelerates the truck + car system. (Static friction from the road on the truck's drive wheels.)

Question 5

`2.4 / 2.5 Net force`

A skydiver opens her parachute and slows from 50 m/s to 6 m/s while still falling. During the slowing, how does the air-drag force compare with her weight?

ASK YOURSELF FIRST

- Which way does her velocity point? Which way does her ACCELERATION point while she slows?
- Net force points the same way as which one, velocity or acceleration?

STEP BY STEP

1. She moves down but slows down, so her acceleration points UP (opposite her velocity).
2. 2nd law: the net force points the same way as a, so it points up.
3. Up – down > 0: drag – mg > 0, so drag > mg.

ANSWER C. Drag is greater than her weight

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Moving down doesn't mean the net force is down. Only speeding up downward would.
- **B:** That's terminal velocity, at the END of the slowing (at 6 m/s).

GO DEEPER Sketch her v–t graph from the moment the chute opens. (It curves toward 6 m/s and levels off, with a steep slope at first.)

Question 6

`2.5 Factor of change`

A net force F gives an object of mass m an acceleration a. What acceleration does a net force 2F give an object of mass 3m?

ASK YOURSELF FIRST

- Write a = F/m. What happens to the top? What happens to the bottom?

STEP BY STEP

1. $a' = (2F)/(3m) = (2/3)(F/m) = (2/3)a$.

ANSWER B. 2a/3

WHY THE WRONG ANSWERS TEMPT YOU

- **C:** Multiplied the two factors. Mass is on the bottom, so you divide by it.
- **D:** Flipped the fraction (F = m/a).

GO DEEPER This 'factor of change' skill (2.D) is about 10–15% of the multiple-choice section. Make it a habit: write the ratio $a'/a = (F'/F)(m/m')$.

Question 7

2.6 Universal gravitation

Two spheres attract each other with force F. The mass of ONE sphere is doubled and the distance between their centers is tripled. The new force is:

ASK YOURSELF FIRST

- How many masses changed? Is distance raised to the first power or the second?

STEP BY STEP

1. $F = Gm_1m_2/r^2$.
2. $F' = G(2m_1)m_2/(3r)^2 = (2/9)F$.

ANSWER B. 2F/9

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Forgot to square the 3.
- **D:** Doubled both masses, but only one was changed.

GO DEEPER What single change to the distance would bring the force back to F? ($r' = \sqrt{2} \cdot r$, if only the one mass is doubled.)

Question 8

2.6 Apparent weight

An elevator's cable snaps and it falls freely (ignore air resistance). A rider stands on a scale inside. During the fall:

ASK YOURSELF FIRST

- Does Earth stop pulling just because the cable broke?
- What's the acceleration of the rider? Plug it into $F_N = m(g - a)$ with down as +.

STEP BY STEP

1. Gravity is unchanged: $F_g = mg$.
2. Rider and elevator both accelerate at g downward. Down +: $mg - F_N = mg \rightarrow F_N = 0$.
3. The scale reads F_N, so it reads zero. This is 'apparent weightlessness', the same thing as on the ISS.

ANSWER B. Scale reads zero; gravity is still mg

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Weight (gravity) and apparent weight (the scale) aren't the same thing.
- **C:** Ignores the acceleration.

GO DEEPER Explain in one sentence why astronauts in orbit and riders in this elevator 'feel' the same. (Both are in free fall, so nothing pushes on them.)

Question 9

2.7 Static vs kinetic

A box sits on a rough floor. A horizontal push starts at zero and increases slowly until well after the box starts to slide. Which graph best shows the friction force vs. the applied force?

ASK YOURSELF FIRST

- Before sliding, how big is static friction compared with your push?
- Once it slides, does kinetic friction depend on how hard you push?
- Which is usually bigger, μ_s or μ_k?

STEP BY STEP

1. At rest: static friction matches the push exactly, so the graph is a line with slope 1 starting at 0.
2. At the maximum $\mu_s F_N$, the box breaks free.
3. Sliding: kinetic friction = $\mu_k F_N$, a constant that doesn't depend on the push. $\mu_k < \mu_s$, so the line drops and then stays flat.
4. That matches graph A.

ANSWER A. Rise along a 45° line, drop, then flat

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** Friction doesn't keep growing. That would mean the box never slides.
- **C:** Friction isn't constant before sliding. With no push, there's no friction at all.
- **D:** Forgets that kinetic friction is smaller than maximum static friction.

GO DEEPER How could you measure μ_s and μ_k from this graph? (The peak ÷ F_N gives μ_s, and the flat part ÷ F_N gives μ_k.) This could be a LAB question.

Question 10

2.7 Incline + kinetic friction

A block slides DOWN a ramp of angle θ at constant speed. The coefficient of kinetic friction must be:

ASK YOURSELF FIRST

- Constant speed means what about the forces along the ramp?
- Write the down-the-ramp push and the friction in symbols, then set them equal.

STEP BY STEP

1. Along the ramp: mg sin θ (down the slope) is balanced by $\mu_k F_N$ (up the slope).
2. Perpendicular: F_N = mg cos θ.
3. mg sin θ = μ_k mg cos θ → μ_k = tan θ. The mass cancels.

ANSWER C. tan θ

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Forgot that the normal force on a ramp is mg cos θ, not mg.

GO DEEPER If the block were twice as heavy, would it still slide at constant speed? (Yes. The mass cancels, so heavier things don't slide faster on the same ramp.)

Question 11

2.9 Flat curve

A car rounds a flat, unbanked curve of radius 50 m. The coefficient of static friction between the tires and road is 0.80. What is the maximum speed without skidding?

ASK YOURSELF FIRST

- Which force points toward the center of the curve? (Only one force can.)
- Is it static or kinetic friction? The tires aren't skidding.

STEP BY STEP

1. Static friction toward the center provides the centripetal acceleration: $f = mv^2/r$.
2. Maximum: $\mu_s mg = mv^2/r \rightarrow v = \sqrt{(\mu_s gr)}$. The mass cancels.
3. $v = \sqrt{(0.80 \times 9.8 \times 50)} = \sqrt{392} \approx 19.8$ m/s.

ANSWER A. 19.8 m/s

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** An extra factor of ½ crept in.
- **C:** Dropped g: $\sqrt{(0.80 \times 50)}$.
- **D:** Forgot the square root (392 is v^2).

GO DEEPER On ice, μ_s drops to 0.20. By what factor does the safe speed drop? ($\sqrt{(0.20/0.80)} = ½$, so it halves.)

Question 12

2.9 Orbits · BOSS

A satellite in a circular orbit of radius r has speed v and period T. It is moved to a circular orbit of radius 4r around the same planet. Its new speed and period are:

ASK YOURSELF FIRST

- What provides the centripetal force? Set that equal to mv^2/r.
- Once you know how v changes, how does $T = 2\pi r/v$ change? Both r and v change.

STEP BY STEP

1. Gravity is the centripetal force: $GMm/r^2 = mv^2/r \rightarrow v = \sqrt{(GM/r)}$.
2. $r \times 4 \rightarrow v \times 1/\sqrt{4} = v/2$. Farther satellites move slower.
3. $T = 2\pi r/v \rightarrow (4r)/(v/2) = 8 \times (r/v)$, so the period becomes 8T.

ANSWER A. v/2 and 8T

WHY THE WRONG ANSWERS TEMPT YOU

- **C:** Assumed farther means faster. It's the opposite.
- **D:** Forgot that the path gets longer AND the speed drops. Both effects increase the period.

GO DEEPER Check it against Kepler's third law, $T^2 \propto r^3$: $8^2 = 64 = 4^3$ ✓.

Question 13

2.8 Springs · optional: ask if springs are on the test

A spring scale is connected between two strings. Each string runs over a pulley and holds a 10 N weight, so the system is at rest. What does the scale read?

ASK YOURSELF FIRST

- If you tied the left string to a wall instead, what would the scale read? Is that any different from what the left weight does?

STEP BY STEP

1. A scale reads the tension in the string attached to it.
2. Each weight hangs at rest, so the tension is 10 N throughout.
3. The left weight does exactly what a wall would do. The scale reads 10 N.

ANSWER B. 10 N

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** 'They cancel' mixes up the net force on the scale (0) with the tension it measures (10 N).
- **C:** Added the two weights together. A scale held by two hands that each pull with 10 N reads 10 N, not 20 N.

GO DEEPER If you used a rope instead, what would its tension be? It's the same idea. (10 N.)

Error log: Quiz B

For each miss or guess: what went wrong, using one of these types: **C** concept (didn't know the physics) · **R** representation (FBD/graph wrong) · **M** math/algebra · **F** FRQ wording (right idea, missing the words) · **T** time.

Q	Type	What I did vs. what I should have done	Re-try date / ✓

Score: _____ / 13 Sure-and-right: _____ Time used: _____